ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION **SUMMER SEMESTER, 2024-2025**

DURATION: 1 HOUR 30 MINUTES **FULL MARKS: 75**

CSE 4803: Graph Theory

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

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1. a) A chemistry teacher has n students in a lab class. Every experiment requires students to work in pairs side-by-side at a long bench arranged in a circle (so each student has exactly one partner on their left and one on their right). The teacher wants to schedule lab sessions such that every student works next to every other student at least once across all sessions, and no two students are seated next to each other in more than one session.

   i. Formulate the scenario as a graph theory problem and determine which concept of graph theory can be used to solve the problem. 7 (CO2) (PO2)

   ii. Assuming the number of students is $n = 7$, labeled A through G, find how many sessions can be scheduled and determine all the possible distinct seating arrangements. $3 + 6$ (CO3) (PO3)

   b) Suppose you are surveying every road in your town to check for damages. The town's road network can be modelled as a graph where intersections are vertices and roads are edges. You want to design a route where you can start from your house, travel all roads exactly once and come back to your house. 9 (CO3) (PO3)

   Assuming that there are 6 intersections with degrees 4, 4, 3, 3, 2, and 2, explain whether it is possible to design such a route with appropriate justifications. If it is possible, construct one such route; and if it is not possible, show which roads need to be used more than once.

2. a) A tech company has 8 employees, each assigned a unique ID number from 1 to 8. The company's internal communication structure during a project is a tree, where every employee is connected to others through a chain of direct one-on-one communication links, with no redundant loops and no employee left out. HR wants to digitally archive every possible such communication structure for compliance purposes.

   However, the archiving system cannot store graphs directly. It can only store short fixed-length numeric codes. An engineer proposes that the communication structures be stored as an $(n - 2)$ size Prüfer code for a graph of n vertices, following the same construction method as Cayley's theorem.

   i. Given the communication tree shown in Figure 1, show the step by step process of converting it into a fixed-length numeric code as suggested by the engineer. 8 (CO1) (PO1)

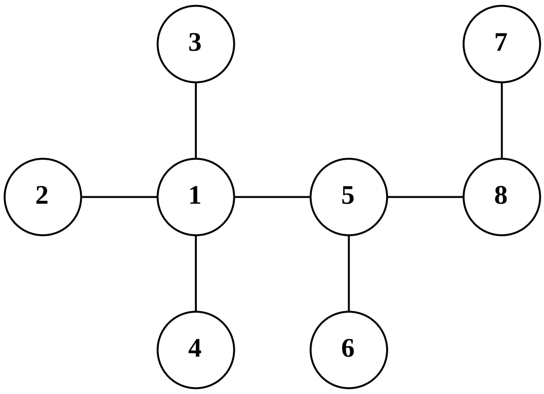

Figure 1: Labeled communication tree for Question 2.a)i

ii. For the next project, the tech company has stored the archive code as $(2, 2, 5, 5, 4, 6)$. Reconstruct the communication tree it represents, showing the step by step process. 8 (CO1) (PO1)

b) Consider the graph G shown in Figure 2 to answer the following questions: $3 + 6$ (CO1) (PO1)

i. Find the maximum possible distance between any two spanning trees of G.

ii. Starting from any spanning tree of G of your choosing, show how you can obtain the spanning tree that is at the maximum distance from your chosen spanning tree.

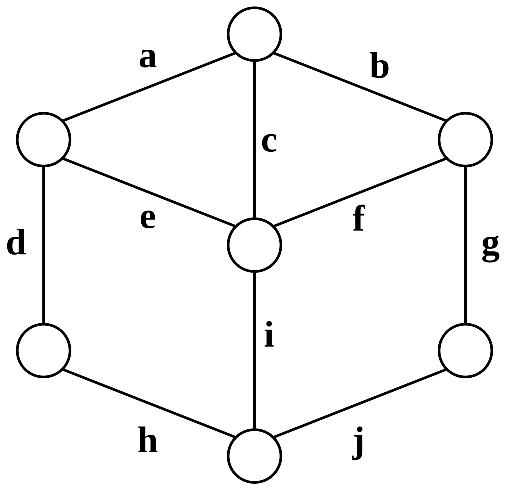

Figure 2: Sample graph G for Question 2.b

3. a) For each pairs of graphs shown in Figure 3 and Figure 4, state whether the two graphs are isomorphic, 1-isomorphic, or 2-isomorphic with appropriate justification, showing the exact criteria you used to test each of them. 2×9 (CO1) (PO1)

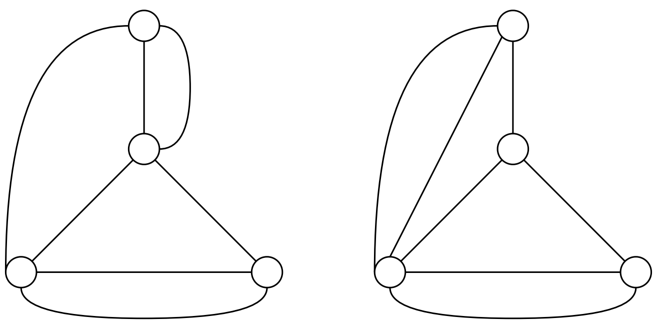

Figure 3: First pair of graphs for Question 3.a

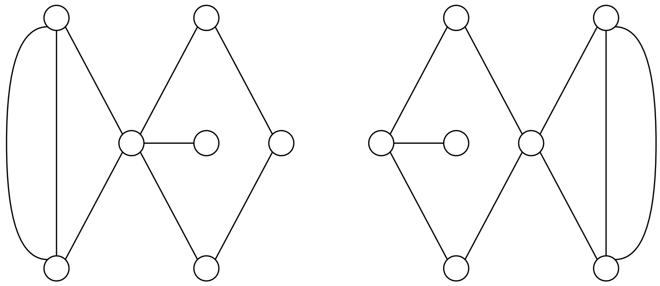

Figure 4: Second pair of graphs for Question 3.a

b) Is it possible for an Euler Graph to have a cut-set with odd number of edges? If it is possible, show one such example. If it is not possible, explain the reasoning. 7 (CO1) (PO1)